# Tetyana Oldham (Konak)

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## Qualification Profile

### Regulatory Knowledge

Graduate Certificate in Regulatory Science expected in May 2019.
Focused on the laws and regulations related to the governance and activities of the US FDA. Course of study covers GCP, GLP, and cGMP, all stages of new drug development process with emphasis ono clinical research. Special attention was focused on the interaction between FDA and pharmaceutical company.

### Quality Control

Developed and validated methods for analyzing and producing products to ensure they meet specifications and document. Created standard operation procedures (SOPs) and production records. Developed quality metrics and assessed trends for product or process implications. Critically evaluated manufacturing processes and identified ways to save time and money without sacrificing product quality. Installed, troubleshooted, and maintain equipment, including calibration, training and repairing when required.

### Laser Physics

Designed new laser set ups based on simple sketch, block-flow, or instrumentation diagrams. Detailed knowledge in theory of optics and laser physics. Knowledge of physical vapor deposition (PVD) and its application to laser physics. Keen understanding of thin films. Synthesized quantum dots and performed optical analyses. Experience in clean rooms. Proven success designing and developing optical materials for lasers. Prepared and characterized pure and transition metal doped II-IV semiconductors. Accomplished at producing new laser materials and building new lasers from the ground up.

### Data Analytics

Developed a new algorithm and tools for data analysis in magnetic resonance imaging (MRI) which simplified the analytic process. Experience with Mathematica, MatLab, Python. Data analysis of pXRD and FTIR spectra. Developing new methods to detect organic and inorganic impurities.

### Mathematics

Research in geometry of submanifolds of Finsler spaces, mean curvature flows, geometry of foliations and distributions, harmonic maps, and standard three-dimensional geometries. Extensive experience with statistics.

### Research

Multidisciplinary research in areas related to lasers, optics, material science, bioimaging, neurobiology and chemical analysis. Ability to adapt to complex and changing situations and technologies and solve cross-fields related research problems.

### Key Strengths

Team player. Excel at building rapport with a wide range of people. Able to convey complex technologies to a variety of skill levels. Ability to analyses ideas and find hidden weaknesses. Talent for quickly learning new information, procedures, and technologies.

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Exceptionally organized and able to multitask. Responsible leader that can delegate as needed. Effective and efficient in group settings.

## Education & Experience

### 2014 - current:

Attending University of Arkansas for Medical Science (UAMS), Department of Public Health, Graduate Certificate in Regulatory Science.

### 2013-2016:

Postdoctoral fellow. National Center for Toxicological Research, U.S. Food and Drug Administration, Jefferson, AR, 72079.

### 2010-2013:

PhD in Physics, The University of Alabama, Birmingham (UAB), AL; Professor Sergey Mirov.

### 2009-2013:

MS in Physics, The University of Alabama, Birmingham (UAB), AL; Professor Sergey Mirov.

### 2002-2003:

MS in Mathematics, Kharkiv National University (KhNU), Kharkiv, Ukraine

## Dissertation:

"Optical and Electrical Characterization of Transition Metal Doped II-VI Mid-IR Laser Materials" under the supervision of Dr. S. Mirov. Research is related to the development of technology for producing new lasing materials, tunable lasers, and providing insight into their optical and electrical properties.

## Experience Highlights

### Quality Control Lead: April 2018-current, Mari Signum Limited, R&D, Richmond, VA.

- Utilizing pXRD, FTIR, UV-Vis, TGA, rheometer, viscometer, titrometer, refractometer for quality control of chitin.
- Developing new methods to detect and quantify common impurities in chitin.
- Developing SOPs, protocols and other quality control documentation.

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Post-Doctoral Fellow: June 2013 – June 2016, Food and Drug Administration, National Center for Toxicological Research, Jefferson, AR.

- Utilized gathered data and techniques to analyze data from MRIs. Developed related software and assisted with research. Utilizing LabView, successfully developed a tool that analyzed MRI images of the brain (in rats) for the purpose of measuring the amount of information on the images and determining the most useful image to use.
- Handling experimental subjects (rats) and assistance in surgeries on them (terminating).

Research Assistant: January 2010 – Jun 2013, University of Alabama at Birmingham, Birmingham, AL.

- Applied cutting-edge knowledge of optics to design, implement, and analyze a new type of laser that was electrically pumped solid state II – IV chromium doped. Utilized thin films, quantum dots, and bulk crystals.
- Developed a more effective way to polish crystals.
- Successfully created new optically active and electrically conductive II – IV materials: Zn:Cr:ZnSe/ZnS, Al:Cr:ZnSe/ZnS.
- Created optically active conductive thin films.
- Developed new quantum dots by modifying standard method which demonstrated high quantum efficiency of pumped conductive samples enhanced with 532 nm laser.

## Publications:

1. Shamshina, J. L.; Oldham, T.; Rogers, R. D. “Chitin Uses in Agriculture,” In: “Applications of Chitin and Chitosan,” G. Crini and E. Lichtfouse (Eds.); Springer/Nature, 2018, under review.

2. Ramu, J.; Konak, T.; Liachenko, Serguei, “Magnetic resonance spectroscopic analysis of neurometabolite changes in the developing rat brain at 7 T,” Brain Research (2016), 1651, 114-120, DOI:10.1016/j.brainres.2016.09.028.

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3. Ramu, Jaivijay; Konak, Tetyana; Paule, Merle G.; Hanig, Joseph P.; Liachenko, Serguei, "Longitudinal diffusion tensor imaging of the rat brain after hexachlorophene exposure," NeuroToxicology (2016), 56, 225-232, DOI:10.1016/j.neuro.2016.08.011.

4. Liachenko, Serguei; Ramu, Jaivijay; Konak, Tetyana; Paule, Merle G.; Hanig, Joseph, "Quantitative assessment of MRI T2 response to kainic acid neurotoxicity in rats in vivo," Toxicological Sciences (2015), 146(1), 183-191, DOI:10.1093/toxsci/kfv083.

5. Hanig, Joseph; Paule, Merle G.; Ramu, Jaivijay; Schmued, Larry; Konak, Tetyana; Chigurupati, Srinivasulu; Slikker, William; Sarkar, Sumit; Liachenko, Serguei, "The use of MRI to assist the section selections for classical pathology assessment of neurotoxicity," Regulatory Toxicology and Pharmacology (2014), 70(3), 641-647, DOI:10.1016/j.yrtph.2014.09.010.

6. Fedorov, V. V.; Konak, T.; Dashdorj, J.; Zvanut, M. E.; Mirov, S. B."Optical and EPR spectroscopy of Zn:Cr:ZnSe and Zn:Fe:ZnSe crystals," Optical Materials (Amsterdam, Netherlands) (2014), 37, 262-266, DOI:10.1016/j.optmat.2014.06.004.

7. Peppers, J.; Konak, T.; Martyshkin, D. V.; Fedorov, V. V.; Mirov, S. B., "Spectroscopy and mid-IR lasing of Cr 2+ ions in ZnSe/ ZnS crystals under visible excitation," Proceedings of SPIE (2014), 8959 (Solid State Lasers XXIII: Technology and Devices), 89590E/1-89590E/7. Language: English, Database: CAPLUS, DOI:10.1117/12.2040798.

8. Konak, Tetyana, "Optical and electrical characterization of transition metal doped II-VI mid-IR laser materials," PhD thesis, (2013), 108 pp..

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9. Konak, Tetyana; Tekavec, Michael; Fedorov, Vladimir V.; Mirov, Sergey B., “Electrical, spectroscopic and laser characterization of γ-irradiated transition metal doped II-VI semiconductors,” Optical Materials Express (2013), 3(6), 777-786, DOI:10.1364/OME.3.000777.

10. Konak, Tetyana; Tekavec, Michael; Fedorov, Vladimir V.; Mirov, Sergey B. “Effects of γ-irradiation on optical, electrical, and laser characteristics of pure and transition metal doped II-VI semiconductors,” Proceedings of SPIE (2011), 7912(Solid State Lasers XX), 791229/1-791229/7, DOI:10.1117/12.875539.